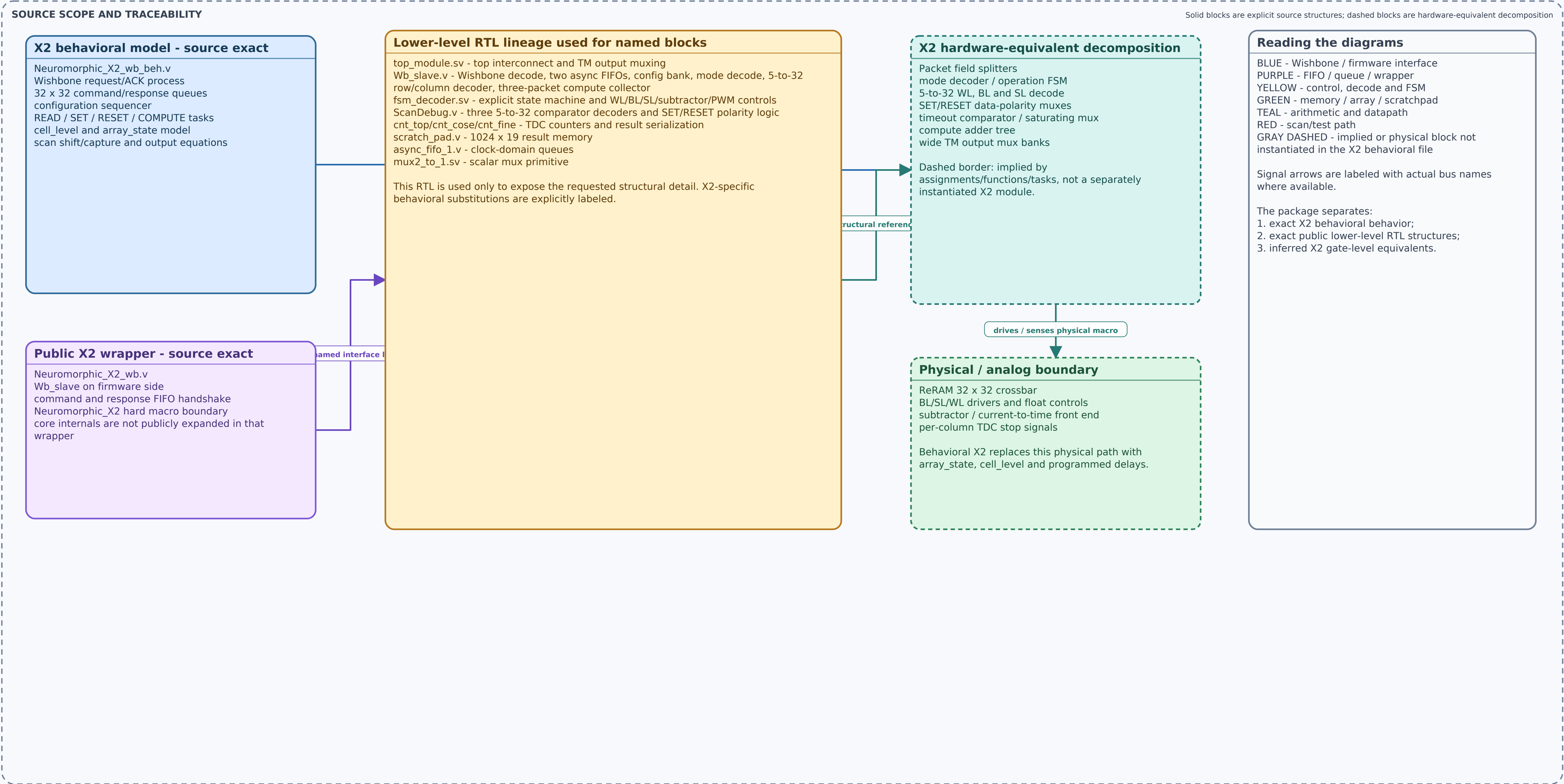


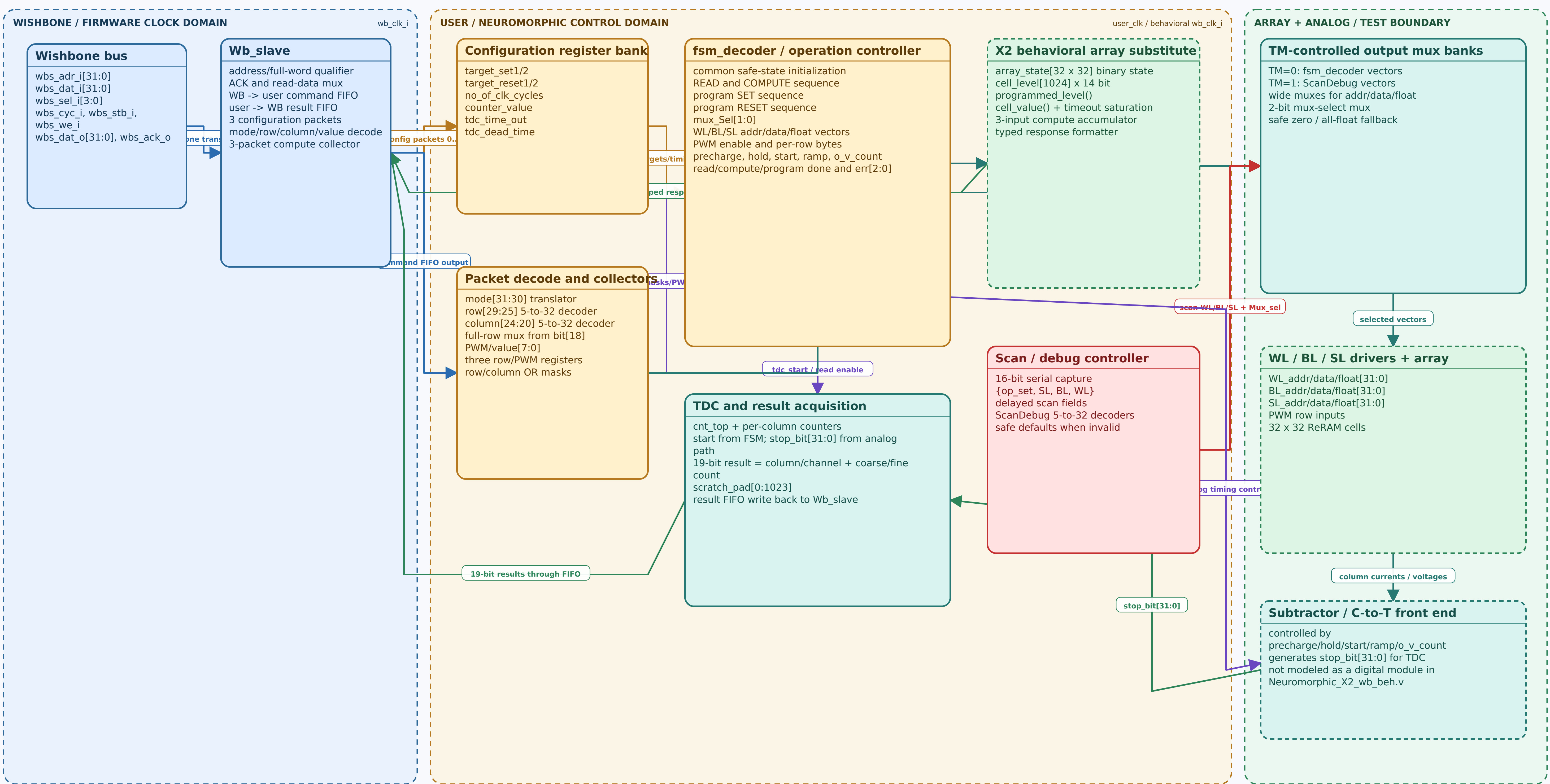
Neuromorphic X2 - Detailed RTL Block-Diagram Scope

How the X2 behavioral source, public wrapper, lower-level RTL lineage, and physical array are combined without hiding inference



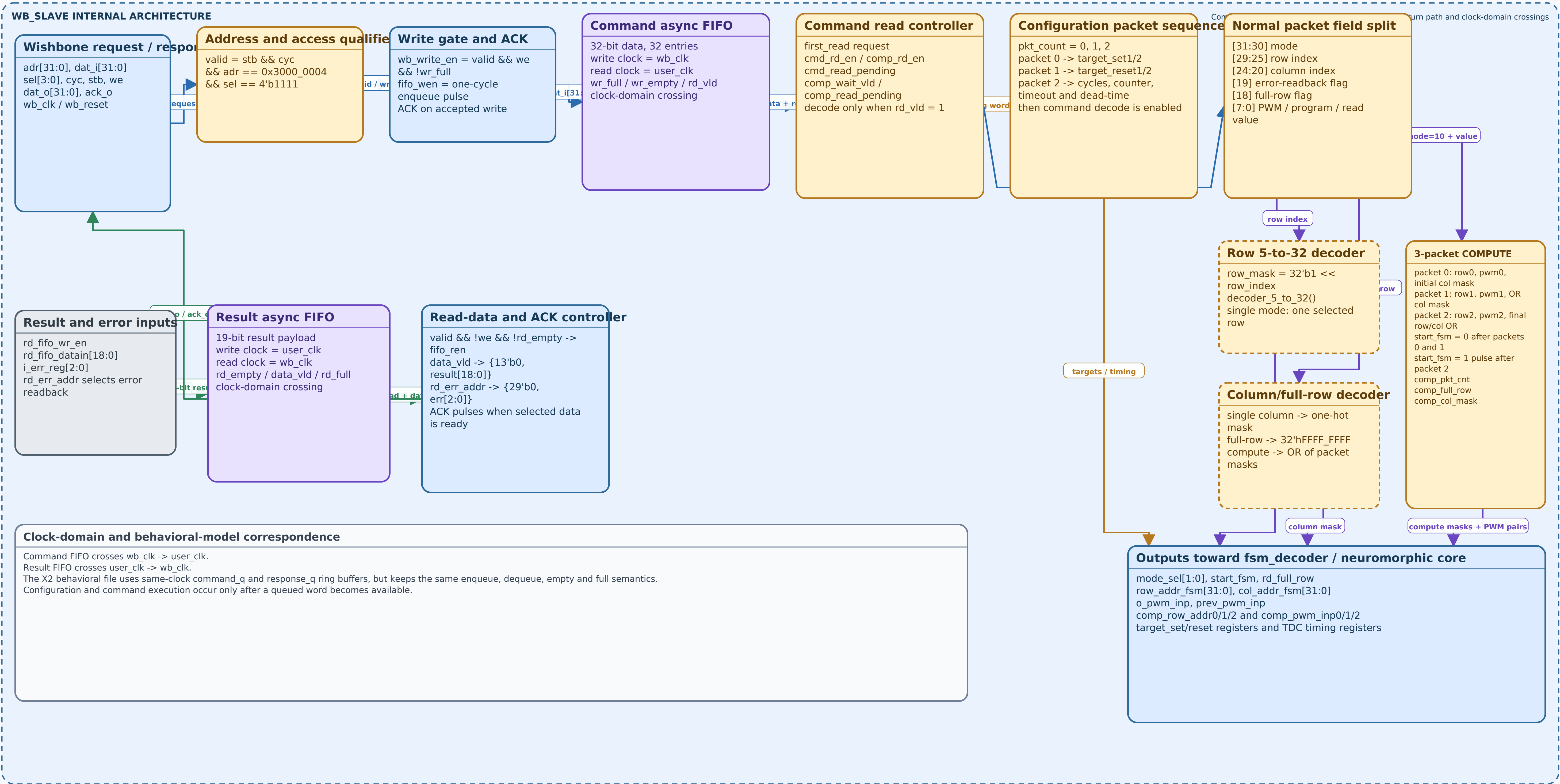
Complete Block-Level Interconnect

Wishbone, command decode, explicit FSM, scan muxing, WL/BL/SL control, subtractor/TDC acquisition, scratchpad and behavioral X2 substitution



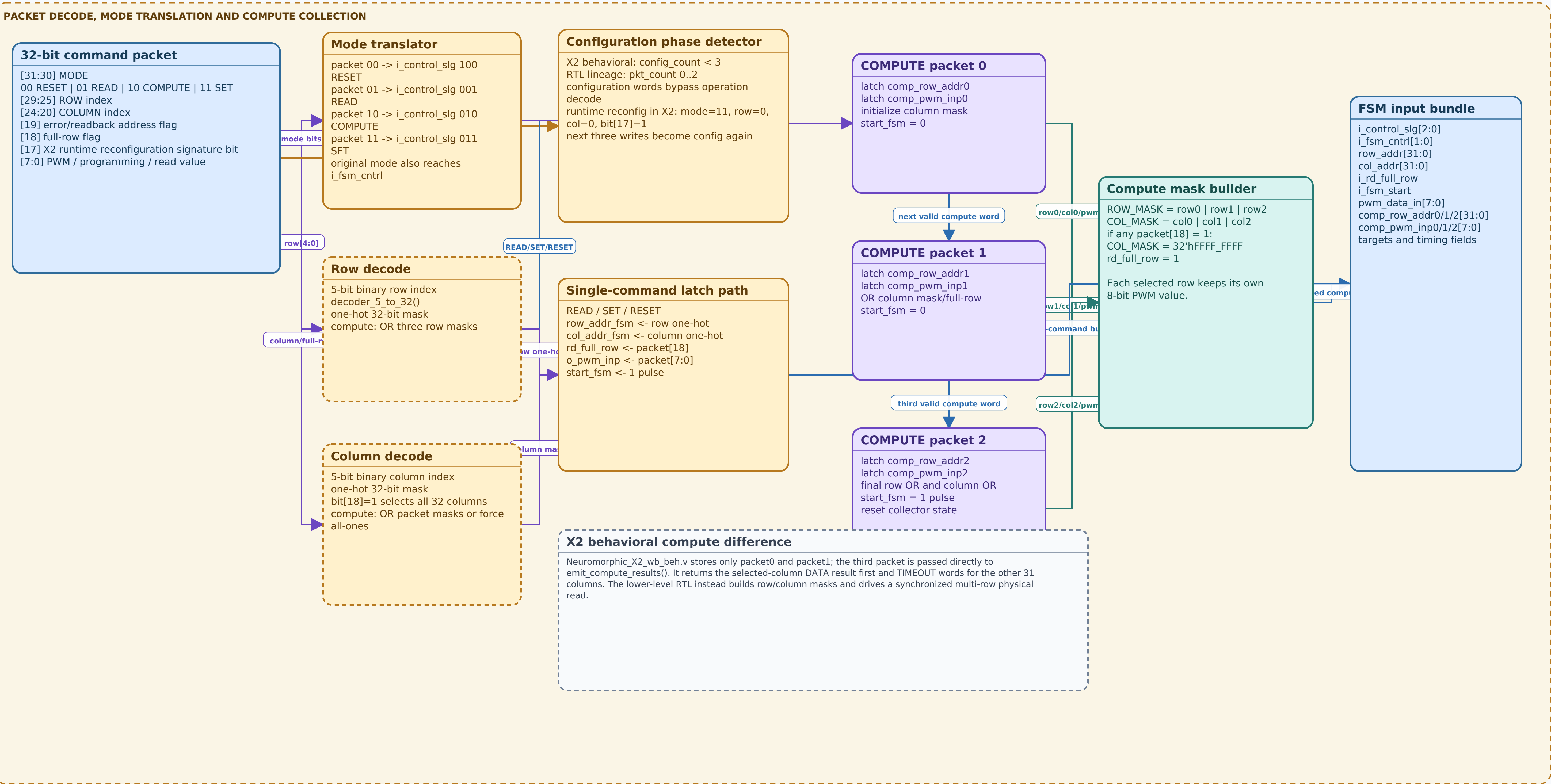
Wishbone Slave - Detailed Internal Blocks

Address compare, ACK logic, two asynchronous FIFOs, configuration sequence, 5-to-32 decoders, compute collection and result/error muxing



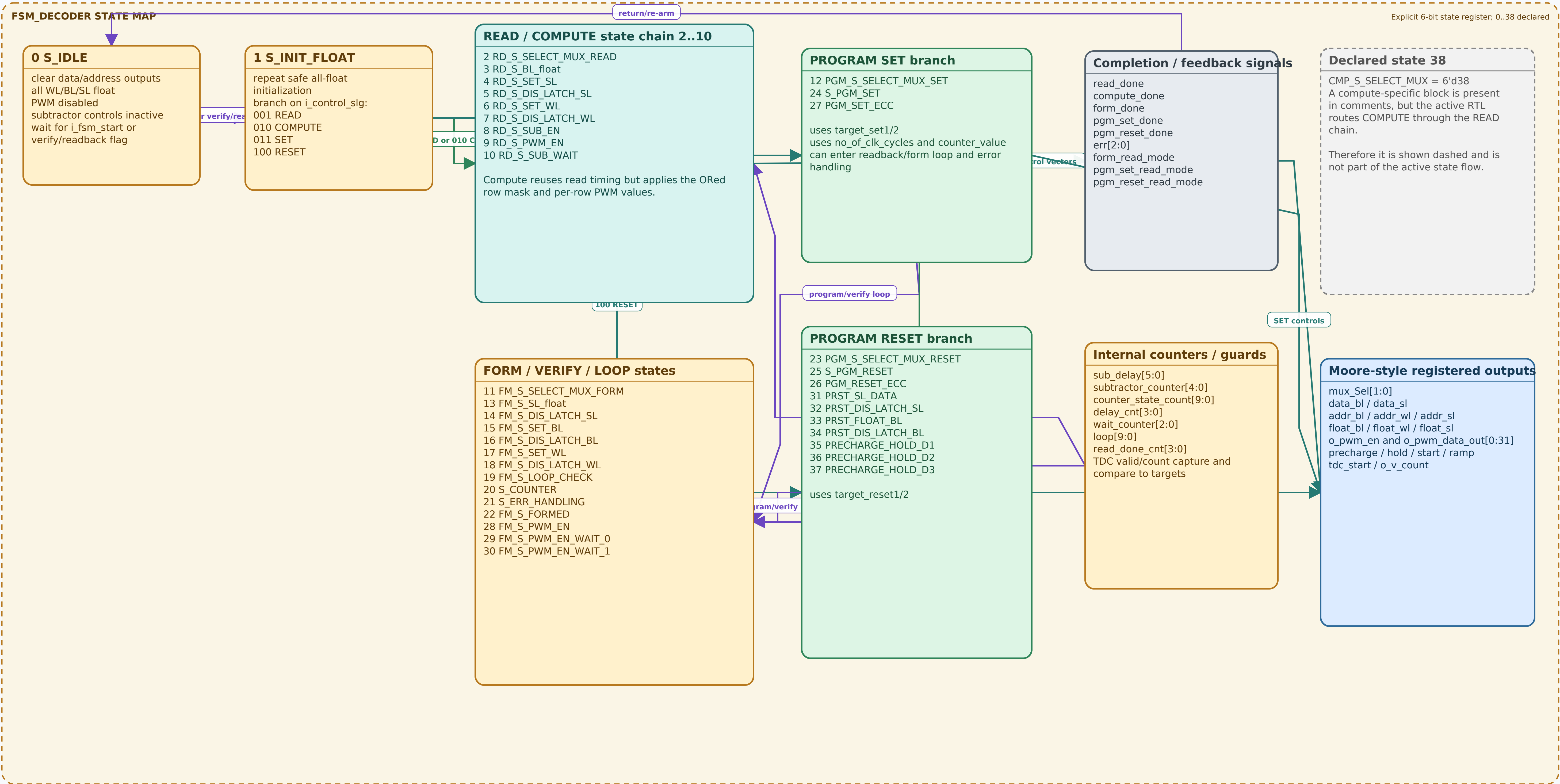
Packet Decoder, Mode Translation and Three-Packet Compute Collector

Every firmware field, one-hot decoder, full-row mux, configuration bypass and synchronized compute aggregation path



fsm_decoder - Full State Map and Mode Branches

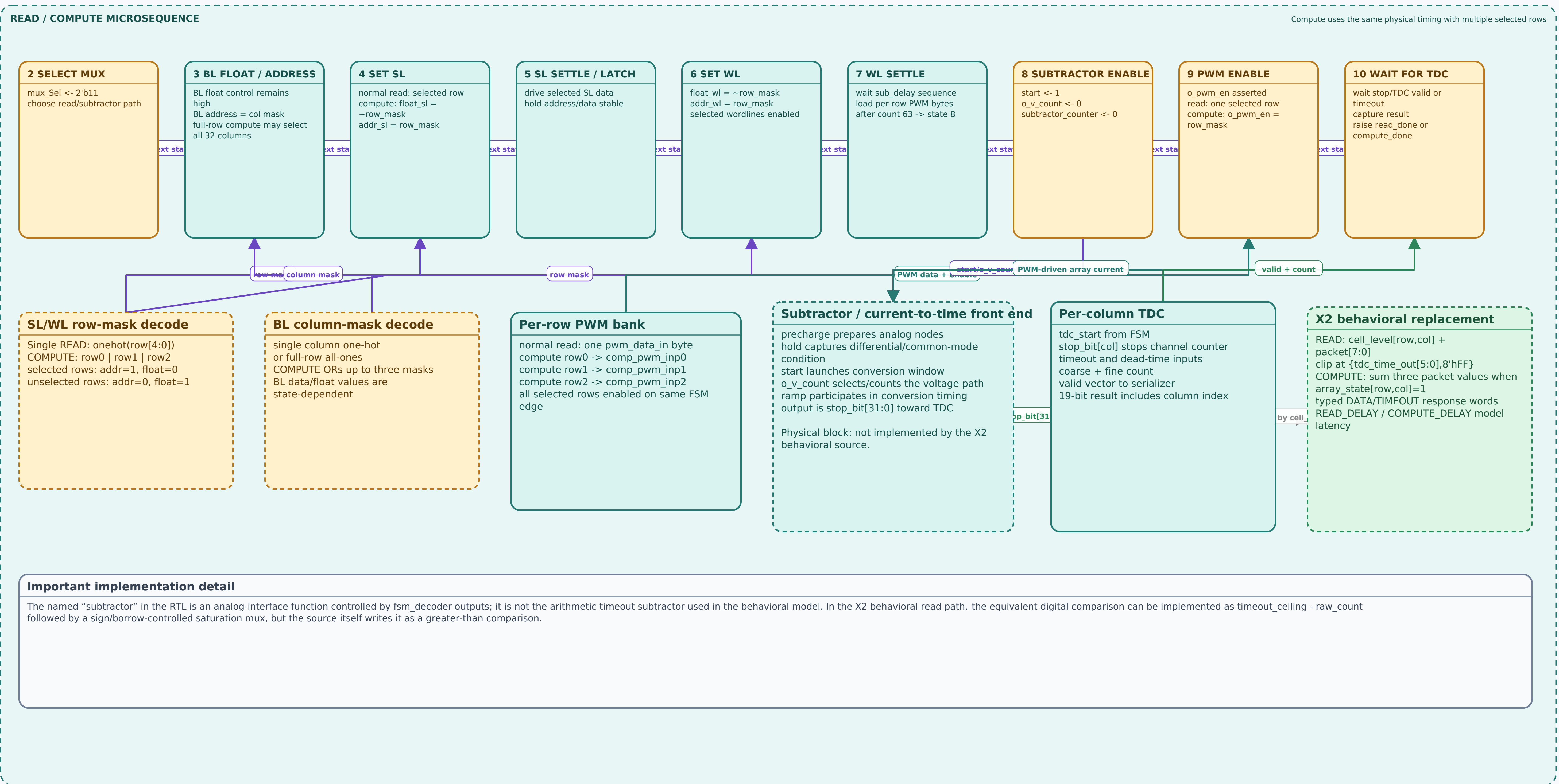
All declared active state groups, control outputs, counters, completion feedback and the commented compute-specific state



READ / COMPUTE - State-by-State Datapath

Mux selection, BL/SL/WL masks, synchronized PWM, subtractor controls, per-column TDC and the exact X2 behavioral abstraction

6 - Read and compute microsequence



PROGRAM SET / RESET - Driver Polarity, Pulse Sequencing and Verify Loops

SET and RESET mux selection, WL/BL/SL polarity, named FSM state groups, target windows, TDC verify and X2 behavioral programming model

PROGRAM SET / RESET AND VERIFY PATHS

PROGRAM SET

packet mode 11 / i_control_slg 011

12 SELECT SET MUX

route programming voltages
load selected row/column masks
use target_set1 / target_set2

SET driver polarity

selected WL data = 1
selected BL data = 1
selected SL data = 0
float = inverse of selected address
PWM/program value from packet[7:0]

24 S_PGM_SET

apply SET pulse sequence
no_of_clk_cycles controls duration
counter_value / loop track
attempts
precharge/hold/PWM states may be reused

SET verify / form loop

readback through TDC
compare measured count with target_set1/2 window
FM_S_* states drive verify read
FM_S_LOOP_CHECK and S_COUNTER decide repeat
S_ERR_HANDLING records failure

27 PGM_SET_ECC and completion

evaluate set-read error condition
raise pgm_set_done
or return to loop/error handling
programmed behavioral X2 state:
array_state[row][col] = 1

PROGRAM RESET

packet mode 00 / i_control_slg 100

23 SELECT RESET MUX

route reset polarity
use target_reset1 / target_reset2

RESET driver polarity

selected WL data = 1
selected BL data = 0
selected SL data = 1
float = inverse of selected address
PWM/program value from packet[7:0]

RESET preparation states 31..37

31 PRST_SL_DATA
32 PRST_DIS_LATCH_SL
33 PRST_FLOAT_BL
34 PRST_DIS_LATCH_BL
35 PRECHARGE_HOLD_D1
36 PRECHARGE_HOLD_D2
37 PRECHARGE_HOLD_D3

25 S_PGM_RESET

apply RESET pulse
duration/attempt counters
drive selected cell with reset polarity
prepare verify readback

RESET verify / form loop

TDC readback
compare count with target_reset1/2 window
loop or error handling
26 PGM_RESET_ECC
raise pgm_reset_done

X2 behavior

array_state = 0
cell_level = midpoint(reset targets) + value
wait PROGRAM_DELAY + no_of_clk_cycles

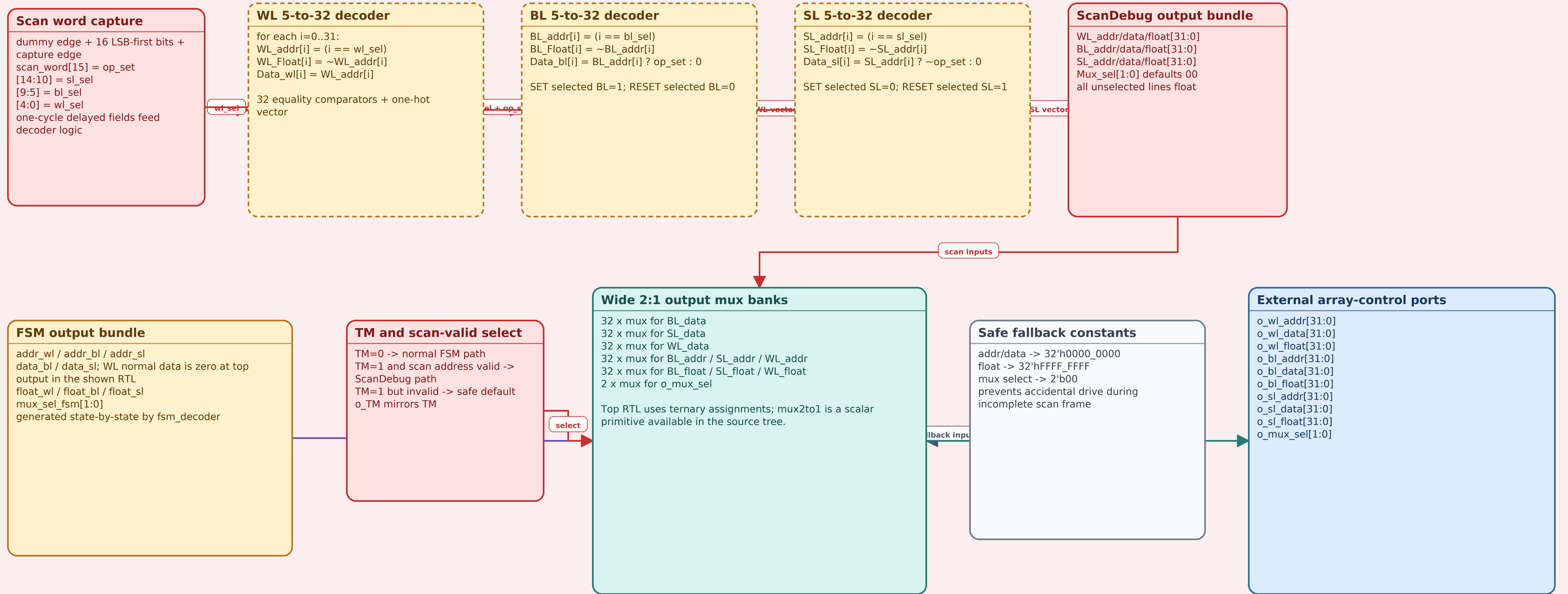
Target midpoint model used by X2 behavioral source

programmed_level = ((min(target1,target2) + max(target1,target2)) >> 1) + zero_extend(packet[7:0]); physical RTL instead sequences pulses and TDC verification around target windows.

Test/Normal Muxes and WL/BL/SL 5-to-32 Decoder Logic

Serial field capture, three one-hot decoder banks, SET/RESET data polarity, wide TM muxes and safe fallback behavior

MUX, WL DECODER, BL DECODER AND SL DECODER DETAIL

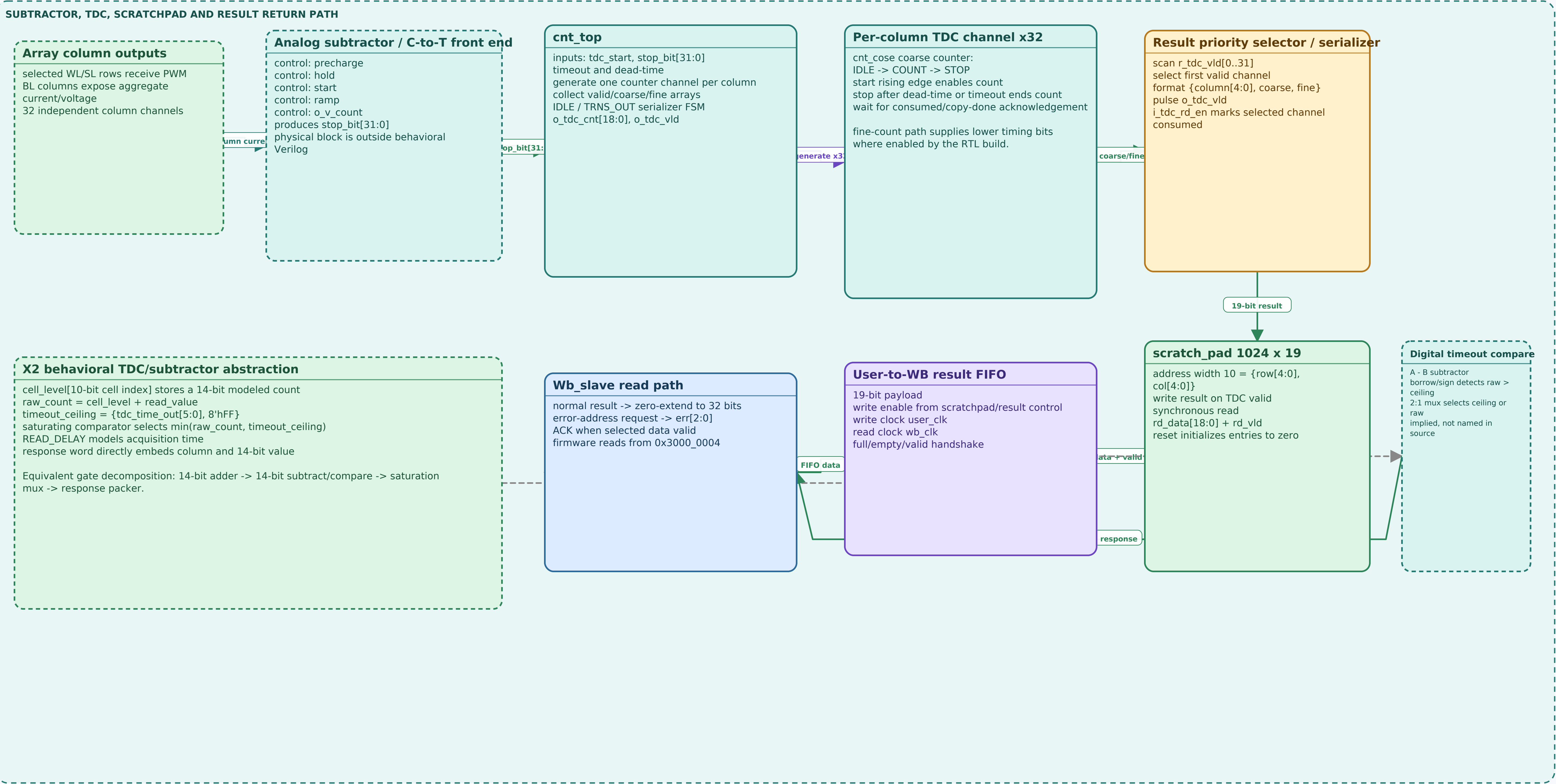


Decoder naming clarification

The public source does not instantiate modules literally named `SL_decoder` or `BL_decoder`. Their logic is inlined as `decoder_5_to_32()` for command masks and as 32 equality comparisons in `ScanDebug/fsm_decoder`. These diagrams expose those combinational blocks explicitly.

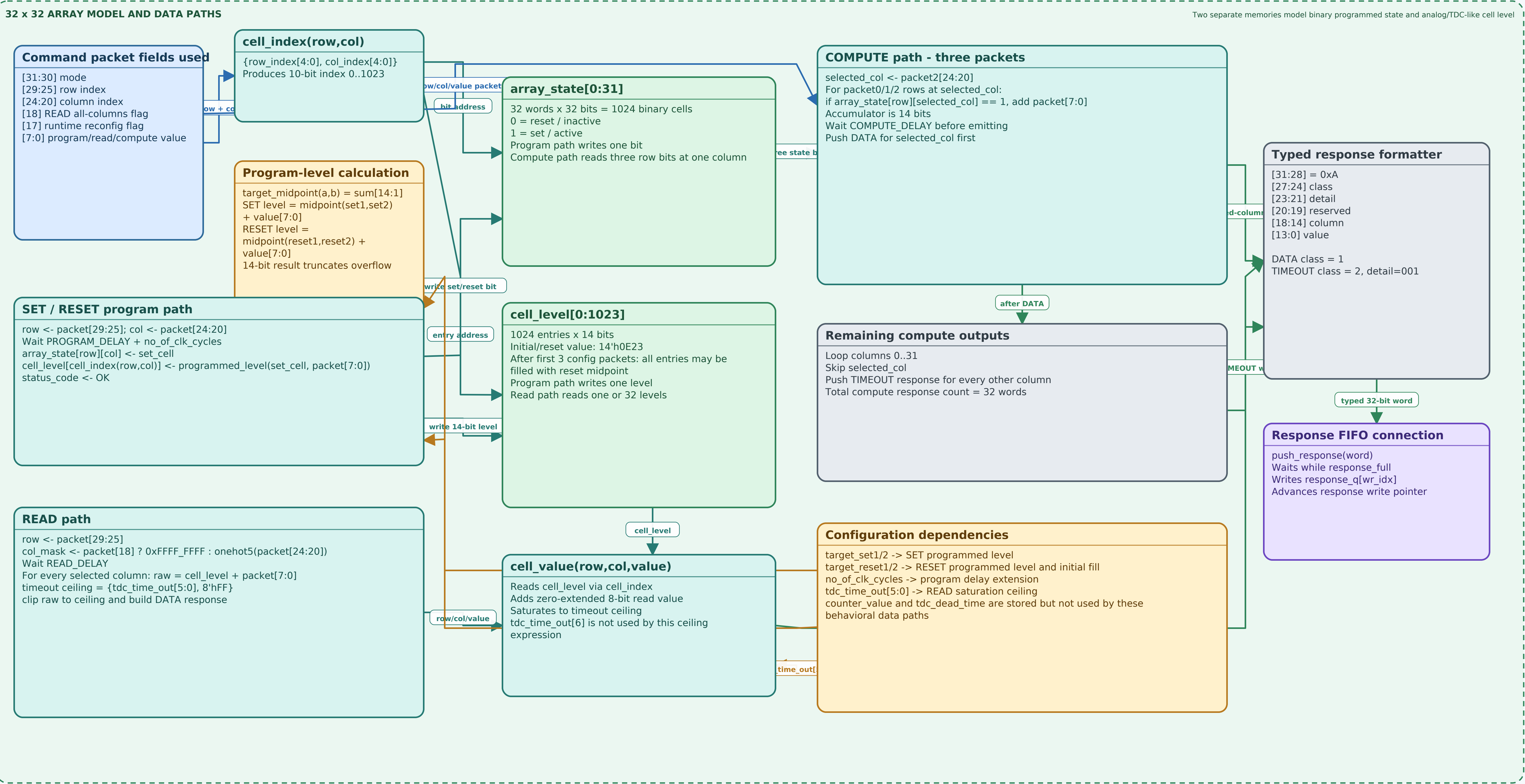
Subtractor / Current-to-Time Front End, Per-Column TDC, Scratchpad and Wishbone Return

Physical analog controls, 32 counter channels, serialization, row/column-addressed result memory, FIFO crossing and the X2 digital timeout equivalent



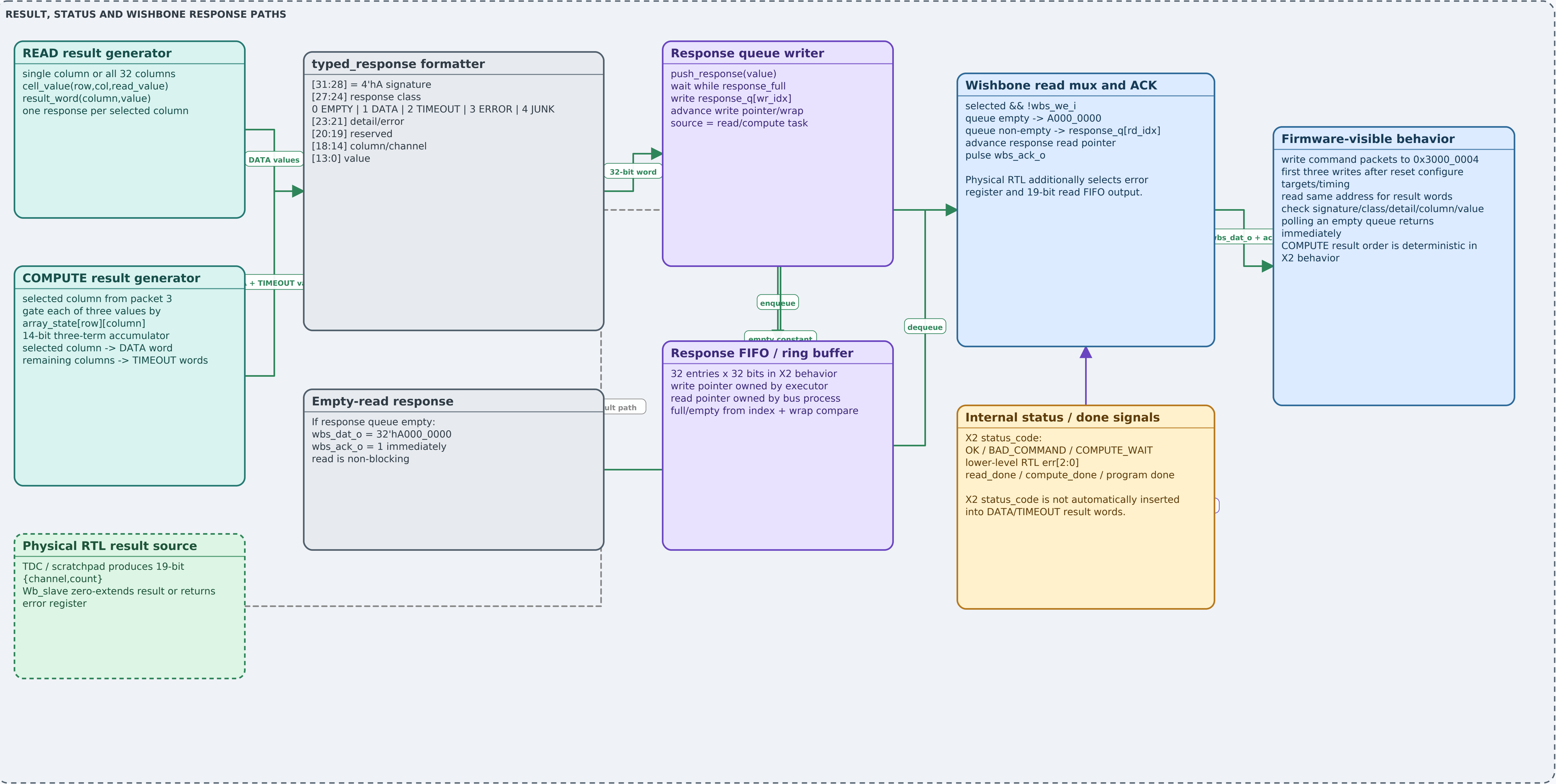
X2 Behavioral Array - Gate-Level Equivalent Datapaths

Programming midpoint arithmetic, 32 x 32 modeled storage, read saturation, compute gating/adder tree and response generation



Response Formatting, Queueing, Status and Wishbone Readback

Typed X2 responses, deterministic compute ordering, empty-read ACK, physical 19-bit result/error mux and firmware decoding



Module-to-Module Port and Signal Matrix

A compact wiring reference for Wb_slave, top_module, fsm_decoder, TDC/scratchpad, ScanDebug/output mux and the physical array boundary

